CORINA KAISER

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Research engineer focused on multi-agent systems, model internals, and visual interfaces.

PROJECTS

Octave RL Environment & Curriculum Controller | Native environment design + staged post-training *2026*

Stack: Python · verifiers v1 (Taskset/Harness) · prime-rl · GRPO · GNU Octave · RTX 6000 Ada

- Built a native verifiers v1 environment for GNU Octave code generation: 10 deterministic task families, 1,500 reference tasks, and 9,000 hidden execution cases for held-out grading.
- Implemented a confidence-gated curriculum controller with policy-consistent static evaluation, durable run state, promotion/demotion logic, and provenance-aware artifact recovery.

Joint-Session | Multi-user BYOK AI workspace *Live Site* · *2026*

Stack: React · Vite · TypeScript · Firebase/Firestore · Cloud Functions · OpenRouter

- Built and deployed a live-sync AI workspace where people and configurable bot personas collaborate concurrently.
- Implemented shared and model-scoped memory with source attribution and cosine retrieval.
- Isolated live in-chat HTML/JavaScript/Python playgrounds for collaborative experimentation.

Scatter Lab | Exploratory data analysis & visualization workbench *Live Site* · *2026*

Stack: TypeScript · Next.js / React · Plotly WebGL · CSV/Excel/Parquet import · Playwright · Vercel (zero backend static deploy)

- Constructed a local analysis core (PCA, DBSCAN, k-means++, three missing-data strategies) and validated it against an independently written implementation, agreeing to eight decimal places.
- Integrated an optional LLM assistant behind a statistical disclosure control layer: only aggregates cross the network, and a per-value small-cell rule (k=5) keeps high-cardinality variables inspectable without exposing individuals.

NVIDIA Nemotron Model Reasoning Challenge *2026*

Objective: Fine-tuning under offline GPU constraints

- Synthesized 9,600 chain-of-thought examples across six symbolic puzzle families and trained rank-32 quantized LoRA adapter for NVIDIA-Nemotron-3-Nano-30B-A3B on RTX Pro 6000 Blackwell in Kaggle's offline environment.
- Patched a Blackwell CUDA toolchain mismatch to complete training under no-network competition constraints.

TECHNICAL METHODS

- **ML / research engineering:** PyTorch, Hugging Face Transformers, PEFT, LoRA/QLoRA, GRPO, evaluation environment design, model routing analysis, control vectors, synthetic data generation
- **Systems Development:** Python, TypeScript, JavaScript, SQL, Node.js, React, FastAPI, Firebase, Supabase, REST APIs, Git, launchd automation
- **Statistics/ data analysis:** PCA, clustering, regression, nonparametric testing, FDR correction, cross-validation, survey-methods analysis, MATLAB, IBM SPSS
- **Hardware:** C++, ESP32-S3, Arduino, CircuitPython, TWAI/CAN, BLE, SMD Soldering, e-Textiles, 3D Printing

EXPERIENCE

LoveSmarter Lab, New York, NY

Research Assistant *Jan 2026–Present*

- Built and reviewed response-quality and bot-detection workflows for large-scale survey data, combining metadata patterns, content coherence, and statistical validation.
- Designed investigations of uncertain-response patterns and subgroup differences using hurdle-style models, FDR-corrected nonparametric tests, partial correlations, cross-validation, and local KNN prediction.
- Developed reproducible cleaning and variable-recoding workflows in Python, MATLAB, and IBM SPSS syntax for research-ready datasets.

SUNY Brooklyn Educational Opportunity Center, Brooklyn, NY

Adjunct Lecturer *May 2023–Jun 2024*

- Developed and delivered adult-learner curricula for Math for Medical Assistants, Math for Food Service Workers, GED Math, and GED Science courses.
- Earlier role: Math & English Tutor and Library Assistant · Dec 2022–May 2023.

EDUCATION

New York University, Tandon School of Engineering · *MS, Mathematical Sciences* *May 2026*

The Pennsylvania State University · *BA, Mathematics* · *BA, Philosophy* · *BS, Applied German* *May 2022*